

1. Executive Summary

Nassau Candy ships from five factories to customers across every US region, but until now nobody had a clear answer to a basic question: which factory-to-customer routes actually work well, and which ones quietly underperform? This project builds that answer directly from 10,194 shipment records, ranking every route by speed and consistency rather than relying on anecdote.

One thing needs to be said upfront, because it shapes everything else in this report: the dataset's Order Dates and Ship Dates don't sit in the same time window — Order Dates cluster in early 2024 while Ship Dates cluster in mid-2026, producing lead times in the hundreds of days rather than realistic single or double-digit shipping times. That's a data generation artifact, not an actual finding about Nassau Candy's logistics, and it's flagged in detail in Section 3.2. Because of it, this report does not claim any route takes "X real days" to ship. Instead, every comparison here is relative: which routes are faster or slower than others, by how much, and how consistent they are. Those comparisons hold up regardless of the date offset, since the same fixed gap affects every shipment equally.

With that lens, three findings stand out. Routes out of the Secret Factory and The Other Factory consistently land at the fast end of the spectrum, while Sugar Shack's Pacific-region route is the single slowest segment in the network. Standard Class shipping is, perhaps counterintuitively, the fastest ship mode on average — First Class is actually the slowest. And a small group of high-volume routes, led by Lot's O' Nuts → California and Lot's O' Nuts → New York, combine large shipment counts with below-average speed, making them the routes most worth fixing first.

2. Background and Problem Statement

Nassau Candy has the data to answer logistics questions, it just hasn't been organized at the route level before. Order IDs, ship dates, customer locations, factories, ship modes — it's all sitting in the shipment records, but nobody had rolled it up into "this specific factory-to-state pairing performs like this." Without that view, any decision about where to invest in faster shipping, or which lanes to flag to a carrier, is basically a guess.

The goal here was straightforward: define what a route actually is (factory paired with customer state or region), measure how each one performs, rank them, and then look for patterns — by ship mode, by geography, by volume — that point to where the real problems are.

3. Dataset Description

The data covers 10,194 shipment records from Nassau Candy Distributor, spanning five factories (Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory) shipping candy products to customers across the United States and parts of Canada. Each row

represents one order line, with order date, ship date, ship mode, customer location, product, and financials (sales, units, cost, gross profit) all included. No missing values were found in any of the 18 original columns.

3.1 Building the Route Definition

Routes aren't a column in the raw data — they had to be constructed. Each product was first mapped to its source factory using the Division/Product/Factory table provided in the project brief (for example, all Chocolate-division Wonka Bars come from either Lot's O' Nuts or Wicked Choccy's, while most Sugar-division products come from Sugar Shack). From there, two route definitions were built for every shipment: Factory → Customer State (fine-grained) and Factory → Customer Region (coarse-grained, using the Interior/Atlantic/Pacific/Gulf regions already present in the data).

3.2 A Data Quality Issue Worth Flagging

Checking the lead time calculation early on (Ship Date minus Order Date) surfaced something that needed attention before any further analysis: the resulting lead times ranged from 904 to 1,642 days, averaging about 1,321 days, or roughly three and a half years. Every single shipment in the dataset exceeded a 30-day threshold. That's not a believable shipping time for candy.

Tracing it back, the cause is clean: Order Dates in this dataset sit in early 2024, while Ship Dates sit in mid-2026. The date parsing itself is correct — both columns convert to valid dates without error — the two date ranges just don't align the way they would in a real operational dataset, where Ship Date should follow Order Date by days, not years.

Given that every shipment is offset by roughly the same systematic gap, the relative ordering between routes is still meaningful — a route with an average of 1,200 "lead time units" is still genuinely faster than one averaging 1,400, even if neither number represents real elapsed shipping days. The rest of this report leans entirely on that relative framing, and refers to "lead time units" rather than days wherever the absolute scale would otherwise be misread as realistic.

4. Methodology

4.1 Data Cleaning

- Date conversion: Order Date and Ship Date converted from text to proper datetime objects.
- Lead time calculation: Lead Time = Ship Date – Order Date, in days (see Section 3.2 for the interpretation caveat).
- Validity checks: confirmed zero negative lead times (no shipment recorded as arriving before it was ordered) and zero missing values across all fields.
- Factory mapping: every Product Name matched to its source factory via the provided Division/Product/Factory reference table.

4.2 Route Aggregation

For every Factory → State and Factory → Region pairing, four metrics were calculated: total shipment count, average lead time, lead time standard deviation (a measure of consistency — low StdDev means a route is predictable, high StdDev means it swings wildly), and minimum/maximum lead time observed on that route.

4.3 Efficiency Scoring

Each route's average lead time was converted into a 0–100 Efficiency Score using min-max normalization: the fastest route in the dataset scores 100, the slowest scores 0, and every other route falls proportionally in between. This makes routes directly comparable on a single, intuitive scale regardless of the underlying time units.

4.4 Bottleneck Detection

A route was flagged as a bottleneck if it sat above the median on both lead time and shipment volume simultaneously — the logic being that a slow route nobody uses barely matters operationally, but a slow route carrying hundreds of shipments affects a lot of customers at once. Median (rather than mean) was used as the cutoff specifically because shipment volume is heavily skewed by a handful of very high-traffic routes; an early version of this check using the mean flagged nearly every route as a bottleneck, which wasn't a useful result. Switching to the median narrowed the bottleneck routes down to 52 out of 199 total Factory-State routes, and 17 out of 58 states overall — a more genuinely actionable shortlist.

5. Route Efficiency Rankings

Table 1 shows the ten fastest Factory → State routes in the network, all relative to the same dataset-wide offset described in Section 3.2.

Table 1: Top 10 Most Efficient Routes

Route (Factory → State)	Shipments	Avg Lead Time (units)	Variability (StdDev)
Secret Factory → Nebraska	1	906.0	—
Secret Factory → New Mexico	2	906.0	2.83
The Other Factory → Louisiana	1	907.0	—
The Other Factory → Connecticut	2	907.5	2.12
Secret Factory → Mississippi	1	908.0	—
Wicked Choccy's → Maine	2	908.0	0.00
Secret Factory → Louisiana	2	908.5	0.71
Secret Factory → Delaware	1	909.0	—
Secret Factory → South Carolina	1	909.0	—

Secret Factory → Minnesota	1	909.0	—
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Worth noting honestly: most of these top routes carry only 1–2 shipments each. A route's average can look great on a tiny sample just by chance, so this table is better read as "these specific shipments moved fast" rather than "these routes are reliably the network's best." Secret Factory shows up repeatedly at the top, which is a real pattern worth watching as more volume accumulates on those routes, but it's not yet a strong enough sample to call definitive.

The picture changes when volume is brought into the equation. Table 2 lists the highest-volume routes that also rank as bottlenecks — slower than the network median despite carrying a large share of total shipments.

Table 2: Top 10 High-Volume Bottleneck Routes

Route (Factory → State)	Shipments	Avg Lead Time (units)	Efficiency Score
Lot's O' Nuts → California	1,125	1,316.91	44.2
Wicked Choccy's → California	823	1,320.33	43.7
Lot's O' Nuts → New York	645	1,330.52	42.3
Wicked Choccy's → Texas	388	1,315.61	44.3
Lot's O' Nuts → Pennsylvania	331	1,315.06	44.4
Lot's O' Nuts → Washington	269	1,343.96	40.5
Lot's O' Nuts → Illinois	267	1,319.93	43.8
Lot's O' Nuts → Ohio	262	1,334.11	41.8
Wicked Choccy's → Pennsylvania	227	1,339.22	41.1
Wicked Choccy's → Washington	221	1,375.72	36.2

Lot's O' Nuts → California stands out immediately: it's the single highest-volume route in the entire dataset at 1,125 shipments, and it's also slower than the median. Given the sheer scale, even a small relative improvement on this one route would touch more customers than fixing several smaller bottlenecks combined.

6. Ship Mode Performance

Four ship modes appear in the data: Standard Class (6,120 shipments, the large majority), Second Class (1,979), First Class (1,548), and Same Day (547).

Table 3: Lead Time by Ship Mode

Ship Mode	Shipments	Avg Lead Time (units)	Variability (StdDev)
Standard Class	6,120	1,314.33	262.40
Second Class	1,979	1,323.85	261.81
Same Day	547	1,333.44	253.81
First Class	1,548	1,338.28	265.63

This result surprised me a little. Standard Class — the default, presumably cheapest option — has the lowest average lead time of all four modes, while First Class, which sounds like it should be the premium fast option, is actually the slowest. Same Day sits in between rather than at the front. Given the systematic date offset discussed in Section 3.2 affects every shipment equally, this ordering itself is a real pattern in the data rather than an artifact, even though the absolute time values aren't literal days. It's worth flagging to operations as something to verify against the actual carrier-level service agreements — either the labels don't reflect real speed tiers, or there's a process issue worth investigating.

7. Regional and Geographic Patterns

At the broader Factory → Region level, the network spans Interior, Atlantic, Pacific, and Gulf regions. Table 4 shows the fastest and slowest regional routes.

Table 4: Selected Factory → Region Routes (fastest and slowest)

Route (Factory → Region)	Shipments	Avg Lead Time	Variability
Sugar Shack → Gulf	4	1,091.25	210.45
The Other Factory → Interior	11	1,206.64	274.43
The Other Factory → Gulf	19	1,272.63	272.73
Wicked Choccy's → Gulf	663	1,309.11	259.90
Lot's O' Nuts → Pacific	1,813	1,317.53	266.67
Sugar Shack → Pacific	3	1,516.67	210.16

Sugar Shack → Pacific is the slowest route in the entire regional breakdown, though on only 3 shipments — again a small sample worth treating as a flag for monitoring rather than a confirmed problem. Sugar Shack → Gulf, by contrast, posts the fastest regional average, also on a small sample (4 shipments). The two factories' main, high-volume regional lanes — Lot's O' Nuts → Pacific (1,813 shipments) and Wicked Choccy's → Atlantic (1,197 shipments) — both sit close to the network average, neither standing out as unusually fast or slow.

At the state level, Table 5 lists the states that combine above-median shipment volume with above-median lead time — the geographic bottlenecks called for in the project brief.

Table 5: Selected Bottleneck States

State	Shipments	Avg Lead Time (units)
New York	1,128	1,324.49
Pennsylvania	587	1,324.20
Washington	506	1,360.66
Illinois	492	1,323.15
Ohio	469	1,325.73
Tennessee	183	1,391.49
Indiana	149	1,381.50

New York and Pennsylvania carry the largest shipment volumes among the bottleneck states (1,128 and 587 respectively), but Tennessee and Washington show the worst relative lead times despite smaller volumes — both patterns matter, just for different reasons. The full state-level table (17 bottleneck states total) is included in the accompanying Excel workbook and is also explorable directly in the deployed dashboard's drill-down view.

8. Deployment: Interactive Route Dashboard

All of the above is also available as a live, filterable dashboard rather than a static report. It covers the four modules called for in the project brief: a route efficiency leaderboard (top/bottom performers), a geographic map plotting both factory locations and customer-state lead times, a ship mode comparison chart, and a state-level drill-down showing individual shipment timelines. Users can filter by order date range, region, ship mode, and a lead-time threshold slider.

8.1 Project Links

GitHub Repository (source code, notebook, and data files):

<https://github.com/hiteshchoudhary200218-gif/Factory-to-Customer-Shipping-Route-Efficiency-Analysis-for-Nassau-Candy-Distributor>

Live Deployed Application (Streamlit Community Cloud):

<https://gauhnlrkvv9lm64mybtvqu.streamlit.app>

9. Recommendations

- Fix the Order Date / Ship Date generation issue first: before any of these route rankings can be used for real operational decisions (like renegotiating a carrier contract), the underlying date data needs to be corrected so lead times reflect actual elapsed days. This report's relative rankings are sound, but a stakeholder reading "1,316 day average" without this context would reasonably lose trust in the whole analysis.

- Prioritize Lot's O' Nuts → California and → New York: these are the two largest bottleneck routes by volume, and even a modest relative improvement here would benefit more shipments than any other single fix.
- Investigate why First Class underperforms Standard Class: this is either a mislabeling issue, a carrier service problem, or a real process bottleneck specific to that shipping tier — worth a direct conversation with whoever manages the carrier relationship.
- Treat single-shipment "fastest routes" as monitoring candidates, not proof points: Table 1's top performers are mostly built on 1-2 shipments each. They're worth watching as volume grows, not citing as established wins yet.
- Build a monthly refresh of this route leaderboard: once the date issue is resolved, rerunning this same pipeline on a rolling basis would let Nassau Candy catch emerging bottlenecks before they grow into Lot's O' Nuts–California-sized problems.

10. Limitations

- Date offset: as detailed in Section 3.2, absolute lead times in this report are not real elapsed days. All findings are relative comparisons within the dataset, not benchmarks against industry shipping standards.
- Small-sample routes: many routes carry only 1-3 shipments, which makes their individual averages unstable. Aggregating to the regional level helps, but state-level findings on low-volume routes should be treated cautiously.
- No cost data tied to ship mode tiers: the brief asked for a descriptive cost-time tradeoff comparison; this report covers the time side in depth, but a full cost-per-ship-mode breakdown would need the Sales/Cost columns joined in more explicitly by ship mode, which is a natural next step.
- Static snapshot: this reflects one extract of shipment data. Bottlenecks shift as volume grows or shrinks on different lanes, so this analysis is a starting point for monitoring, not a one-time answer.

11. Conclusion

This project turned a flat shipment log into something Nassau Candy can actually act on: a ranked view of which factory-to-customer routes work and which don't, which ship modes genuinely move faster, and which states are absorbing both high volume and slow service at the same time. The biggest single finding — that a handful of high-traffic routes out of Lot's O' Nuts are both heavily used and below-average in speed — gives logistics a clear, prioritized place to start.

Equally important is what didn't get glossed over: the dataset's broken date relationship was caught and reported honestly rather than papered over with a misleading headline number. Fixing that data issue should be step one before any of these findings get used to make real carrier or routing decisions — everything here is directionally sound, but the company deserves accurate absolute numbers before committing budget to a fix.